FINANCIAL AI EVALUATION

Case Study 01
Financial Expectancy Evaluation

Domain: Financial AI Evaluation | Category: Quantitative Reasoning | Evaluation: Mathematical accuracy

Dataset	100 hypothetical trades	**Verdict**	Incorrect conclusion
Win rate	45%	**Reward / risk**	2R / 1R

1. Evaluation objective

Assess whether an AI system correctly interprets trading profitability when the strategy's win rate is below 50% but its average winning trade is larger than its average losing trade.

2. Scenario and AI-generated claim

Input data: 100 trades; 45 winners; 55 losers; average winning trade +2R; average losing trade –1R.

> *"The strategy has a 45% win rate, which means it is unprofitable because the win rate is below 50%."*

3. Independent calculation

Expectancy = (Win rate × Average win) – (Loss rate × Average loss magnitude)

E = (0.45 × 2R) – (0.55 × 1R) = **+0.35R per trade**

Expected aggregate result over 100 trades = 100 × 0.35R = +35R

The calculation is a gross expectancy estimate under the stated assumptions. It excludes commissions, spread, slippage, financing costs, and execution differences.

4. Identified reasoning errors

Win-rate misinterpretation: The AI assumes that a strategy needs a win rate above 50% to be profitable.

Ignoring reward-to-risk: It fails to account for the size of average wins relative to average losses.

Unsupported conclusion: It labels the strategy unprofitable without calculating expectancy or considering costs.

5. Corrected assessment

The supplied figures imply positive gross expectancy of +0.35R per trade. Therefore, a 45% win rate alone does not justify calling the strategy unprofitable. A complete assessment must also account for trading costs and verify that the stated average outcomes are based on a representative sample.

6. Evaluation conclusion

Verdict: Incorrect. The AI's conclusion is mathematically unsupported. It relies on win rate in isolation and ignores expectancy and reward-to-risk. The corrected gross expectancy is +0.35R per trade, subject to the assumptions and exclusions noted above.

Portfolio transparency: This is an independently completed educational evaluation using a hypothetical trading scenario. It is not a paid client project or a record of live trading performance.

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CASE STUDY 02 | OVERFITTING & DATA SNOOPING

Evaluation category: Statistical Reasoning & Backtesting Reliability

1. Scenario

An evaluator is given a backtesting summary for 50 trading strategies tested on the same historical dataset. The best-performing strategy reports an 82% win rate, a 3.4 profit factor, 120 trades, and a 4% maximum drawdown. An AI system concludes that the strategy is likely to remain profitable in live trading.

2. Core evaluation question

Does the strongest result among 50 strategies provide reliable evidence of future profitability, or could the result be inflated by repeated experimentation and selection?

3. Evaluation findings

Overfitting: A strategy may fit historical noise rather than persistent market structure. Its impressive historical metrics may not generalize to unseen data.

Data snooping / multiple testing: Testing 50 strategies on the same dataset creates repeated opportunities to find a strong result by chance. Selecting only the best result introduces selection bias and can exaggerate expected performance.

Unsupported live-performance inference: A high historical win rate, profit factor, and low drawdown do not by themselves establish that the strategy will work in live markets.

4. Recommended validation process

Stage	Purpose
Training / development data	Develop and tune strategies; do not treat these results as final evidence.
Validation data	Compare candidates and make limited design choices without repeatedly reusing the final test set.
Untouched test data	Evaluate the selected strategy once on genuinely unseen, chronologically later data.
Walk-forward testing	Repeat development and evaluation across sequential time windows to test stability.
Forward / paper testing	Observe performance on new market data before risking capital.

5. Additional checks before trusting the result

• Include realistic commissions, spread, slippage, funding and execution assumptions.
• Check whether the 120 trades are sufficiently independent and representative; clustered or overlapping trades can make evidence look stronger than it is.
• Review performance across different market regimes and report the full distribution of tested strategies, not only the winner.
• Keep a record of how many variants, parameter combinations and repeated experiments were tried.

6. Corrected AI assessment

The reported backtest is promising as an exploratory result, but it is not sufficient to conclude that the strategy is likely to remain profitable live. Because 50 strategies were tested on the same historical dataset

and only the best result was highlighted, the reported metrics may be affected by overfitting, data snooping and selection bias. The strategy should be assessed using a pre-defined validation process, an untouched chronological test period, walk-forward analysis, realistic trading costs and forward testing.

7. Evaluation conclusion

AI verdict: Incorrect / overconfident. The AI treats the best in-sample result as strong evidence of live profitability without accounting for multiple testing or out-of-sample validation. The appropriate conclusion is that the strategy requires further testing.

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CASE STUDY 03 | CONDITIONAL PROBABILITY

Evaluation category: Probability, Statistical Reasoning & Causal Interpretation

1. Scenario

A trading dataset contains the following observations: 78% of winning trades occurred after a liquidity sweep, while 32% of losing trades also occurred after a liquidity sweep. An AI system interprets these figures as proof that a liquidity sweep makes a trade highly likely to win.

2. Core evaluation question

Do the supplied percentages establish the probability of winning after a liquidity sweep? Or has the AI reversed the conditional probability?

3. Key probability distinction

P(Sweep | Win) = 78% means that 78% of winning trades had a liquidity sweep beforehand. It describes the frequency of sweeps among winning trades.

P(Sweep | Loss) = 32% means that 32% of losing trades had a liquidity sweep beforehand. It describes the frequency of sweeps among losing trades.

P(Win | Sweep) is a different quantity: the probability that a trade wins given that a liquidity sweep occurred. This is the conditional probability needed to assess the win likelihood after a sweep.

4. Why the AI conclusion is unsupported

The AI has treated P(Sweep | Win) as if it were P(Win | Sweep). These probabilities are not interchangeable. The supplied percentages alone do not reveal how often sweeps occur overall, how many trades win or lose, or the win rate among trades that actually had a sweep.

5. What information is required?

Required information	Why it matters
Total number of trades	Establishes the sample size and context.
Number of winning and losing trades	Provides the base rates for wins and losses.
Number of trades with a sweep	Shows how common the condition is.
Wins among sweep trades	Allows calculation of P(Win | Sweep).
Comparable no-sweep results	Helps assess whether the condition adds useful predictive information.

6. Further analytical checks

• Compare P(Win | Sweep) with P(Win | No Sweep), using the same clearly defined dataset.
• Check sample size and uncertainty; small samples can produce unstable percentages.
• Consider confounding variables such as market regime, volatility, session, trend direction and entry confirmation.
• Test the relationship on unseen data before treating it as a reliable trading signal.
• Even a predictive association does not prove that the sweep causes the trade to win.

7. Corrected AI assessment

The available figures show that liquidity sweeps were more frequently observed among winning trades than among losing trades in this sample. However, they do not directly establish the probability of a win after a sweep. To evaluate that question, calculate P(Win | Sweep) from the underlying counts, compare it with the no-sweep baseline, assess uncertainty and possible confounders, and validate the result out of sample.

8. Evaluation conclusion

AI verdict: Incorrect — conditional probability reversal. The AI confuses P(Sweep | Win) with P(Win | Sweep) and makes an unsupported predictive and causal claim. The correct conclusion is that the supplied figures suggest an association worth testing, but are insufficient to quantify the probability of winning after a sweep.

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CASE STUDY 04 | DATA LEAKAGE & LOOK-AHEAD BIAS

Evaluation category: Machine Learning Validation & Time-Series Integrity

1. Scenario

A model is trained to predict BTC price direction using 10,000 historical candles. The dataset is randomly divided into training and test subsets. Before the split, a normalization scaler is fitted using the entire dataset. The model reports 91% training accuracy and 89% test accuracy. An AI evaluator concludes that the model generalizes well because the two accuracy scores are close.

2. Core evaluation question

Can the reported test accuracy be trusted when preprocessing was fitted on the full dataset and a time-series dataset was randomly split?

3. Identified issues

Preprocessing leakage: The scaler learned information from the full dataset, including observations assigned to the test set. The test data therefore influenced preprocessing before evaluation. This can make test performance an optimistic estimate.

Time-series split concern: Randomly splitting sequential candle data can place later observations in training and earlier observations in testing. This may fail to reflect the real forecasting task, where a model is trained on past data and evaluated on future data. Whether it causes direct target leakage also depends on feature construction, labels and overlapping observation windows.

Overconfident interpretation: Similar training and test accuracy does not prove generalization when the evaluation design itself is contaminated or unrealistic.

4. Correct validation workflow

Step	Correct procedure
1. Chronological split	Separate older observations for training, later observations for validation and the latest period for final testing.
2. Fit preprocessing	Fit scalers and other learned transformations on training data only.
3. Transform later sets	Apply the already-fitted training transformations to validation and test data without refitting.
4. Tune carefully	Use validation data for model selection; keep the final test period untouched until the final evaluation.
5. Walk-forward test	Repeat the process across sequential windows to assess stability over time.

5. Additional checks

• Confirm that all features are available at the actual prediction time; remove features that contain future information.
• Check label construction and overlapping look-ahead horizons. Where necessary, use purging or an embargo between training and test observations.
• Evaluate more than accuracy: include class balance, precision, recall, F1, calibration and financial metrics such as net expectancy after realistic costs.
• Compare results with simple baselines and inspect performance across market regimes.

6. Corrected AI assessment

The 91% training accuracy and 89% test accuracy cannot establish reliable generalization because the scaler was fitted on the complete dataset before the split. This introduces preprocessing leakage. In addition, a random split is generally unsuitable for evaluating future BTC predictions unless the design specifically prevents temporal contamination and respects the forecasting setup. Rebuild the pipeline using chronological splits, fit all learned preprocessing steps on training data only, and evaluate on untouched future data with walk-forward validation.

7. Evaluation conclusion

AI verdict: Incorrect / insufficiently supported. The evaluator overlooks leakage from full-dataset normalization and does not challenge the random split for a sequential forecasting problem. The reported accuracy should not be accepted as reliable evidence until the validation pipeline is corrected.

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CASE STUDY 05 | SURVIVORSHIP BIAS

Evaluation category: Dataset Integrity, Selection Bias & Performance Reporting

1. Scenario

A research report evaluates 100 trading strategies that were launched over a historical period. Only the 15 strategies that survived and remained profitable are included in the final performance summary. The report states that the average win rate of these 15 strategies is 68% and concludes that the strategy universe is generally highly effective.

2. Core evaluation question

Can the performance of the 15 surviving strategies represent the original group of 100, when the 85 excluded strategies may have failed, underperformed or been discontinued?

3. Identified evaluation issues

Survivorship bias: The analysis excludes strategies that did not survive. By observing only the winners or survivors, the report creates an overly positive picture of historical performance.

Selection bias: The 15 strategies were selected based on their outcome. Their performance is therefore not an unbiased estimate of what would have been expected from the original group of 100 strategies.

Incomplete reporting: Reporting only the survivors hides the failure rate, distribution of outcomes, losses and the criteria used to exclude strategies.

4. Why the reported 68% average is misleading

The 68% win rate describes the selected subset of 15 surviving strategies, not the full population of 100. It cannot be generalized to the original strategy universe without including the excluded strategies and understanding the selection process. The 85 excluded strategies must not automatically be assigned a 0% win rate; their actual records should be recovered and evaluated.

5. Recommended evaluation framework

Requirement	Evaluation purpose
Complete strategy universe	Include all 100 strategies, including discontinued and losing ones.
Consistent inclusion rules	Define eligibility and exclusion criteria before examining outcomes.
Full outcome distribution	Report winners, losers, discontinued strategies, returns and drawdowns.
Selection process disclosure	Explain when and why the 15 strategies were chosen.
Out-of-sample validation	Test whether conclusions hold on unseen data, not just the period used for selection.
Multiple-testing awareness	Account for repeated strategy searches and parameter experiments where relevant.

6. Additional checks

• Verify that every strategy in the original group is accounted for, including those with missing records or early termination.
• Avoid calculating a single average without showing dispersion, median, worst outcomes and failure rates.

• Ensure that all strategies are measured over comparable periods and with consistent transaction-cost assumptions.
• Distinguish survivorship bias from related issues such as data snooping and repeated selection of the best-performing strategy.

7. Corrected AI assessment

The 68% average win rate applies only to the 15 selected surviving strategies. Since 85 of the original 100 strategies are excluded, the report is affected by survivorship and selection bias. The figure cannot support a conclusion that the broader strategy universe is generally effective. A reliable evaluation requires the complete original set, transparent exclusion rules, consistent performance measurement and validation on data not used for selection.

8. Evaluation conclusion

AI verdict: Incorrect / materially biased. The report generalizes from a selected survivor subset to the full strategy universe while omitting most of the original sample. The correct conclusion is that the 15 survivors performed well on the reported measure, but the broader effectiveness of the 100-strategy group remains unknown.

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CASE STUDY 06 | COMPREHENSIVE MODEL EVALUATION

Evaluation category: Quantitative Accuracy, Backtesting Integrity, Statistical Validity & Risk Assessment

1. Scenario

An AI-generated research report evaluates a BTC trading strategy using 3,000 historical trades. It reports a 58% win rate, average winning trade of +1.5R, average losing trade of –1R, gross expectancy of +0.45R per trade, transaction costs of 0.12R per trade, profit factor of 1.65 and maximum drawdown of 12%. The strategy was selected as the best of 80 candidates. The entire dataset was normalized before a random train/test split.

2. Executive assessment

The report contains a promising headline result, but its reliability is not established. The net expectancy can be recalculated from the stated win/loss assumptions, while the reported profit factor should be reconciled with the implied gross figures. The selection of one strategy from 80 candidates and preprocessing before the split raise important validation concerns.

3. Quantitative verification

Metric	Independent calculation	Assessment
Win rate	58% (given)	Input metric; needs a clear trade definition.
Gross expectancy	(0.58 × 1.5R) – (0.42 × 1R) = +0.45R	Matches the report.
Net expectancy	+0.45R – 0.12R = +0.33R/trade	Correct under the stated average-cost assumption.
Expected aggregate net result	3,000 × 0.33R = +990R	Conditional arithmetic, not a realized or guaranteed result.
Implied gross profit factor	(0.58 × 1.5) ÷ (0.42 × 1) = 2.07	Does not match the reported PF of 1.65; reconcile definitions/data.
Maximum drawdown	12% (reported)	Cannot be independently verified without the equity curve and calculation

4. Critical methodological findings

A. Data leakage: Normalizing the full dataset before the train/test split allows test-set information to influence preprocessing. Learned transformations must be fitted on training data only and then applied unchanged to later data.

B. Random time-series split: For a strategy intended to predict future BTC prices, a random split may mix earlier and later observations across training and testing. Use chronological splits and walk-forward validation. Check overlapping labels and feature windows; random splitting is not automatically proof of direct look-ahead leakage in every possible dataset, but it is generally an unsuitable estimate of future deployment performance.

C. Multiple testing / selection bias: The strategy was selected as the best among 80 candidates. The best observed result may be inflated by repeated experimentation. Record all candidates and tuning decisions, and evaluate the selected strategy on a genuinely untouched test period.

D. Profit-factor inconsistency: The stated win rate and average win/loss imply a gross profit factor of approximately 2.07, not 1.65, if all figures use the same trade set and R definition. The discrepancy could reflect different aggregation methods, trade subsets, position sizing or costs, but the report must explain it rather than leave it unresolved.

E. Risk metric limitations: A 12% maximum drawdown is not enough to characterize risk by itself. The report should disclose the equity curve, drawdown duration, exposure, position sizing, tail losses, leverage and the method used to calculate drawdown.

5. Required remediation and validation plan

Priority	Required action
1 — Rebuild pipeline	Create chronological train, validation and untouched test periods; fit normalization only on training data.
2 — Audit experiments	Document all 80 strategies, parameter searches and selection criteria; account for multiple testing.
3 — Reconcile metrics	Recalculate PF from raw trade-level gross profit/loss and explain the difference from 2.07.
4 — Recompute net results	Apply realistic fees, spread, slippage, funding and execution assumptions consistently.
5 — Validate stability	Run walk-forward tests across market regimes and compare with simple baselines.
6 — Forward test	Paper trade on new data and monitor drift before considering live deployment.

6. Corrected AI assessment

The strategy shows positive stated net expectancy of +0.33R per trade after the assumed 0.12R cost, and the arithmetic implies +990R over 3,000 trades if those averages apply consistently. However, this is a conditional calculation, not proof of future profitability. The implied gross profit factor of approximately 2.07 conflicts with the reported 1.65 and requires reconciliation. In addition, selecting the best of 80 strategies, fitting normalization on the full dataset, and using a random split weaken the credibility of the reported performance. The model should not be considered validated until these issues are resolved using leakage-free chronological and walk-forward testing.

7. Final evaluation verdict

AI verdict: Not validated — significant methodological and reporting issues. The headline expectancy is arithmetically positive under the stated assumptions, but performance reliability remains unproven. The PF discrepancy, data leakage, random split and selection among 80 candidates must be addressed before any claim of robust or live profitability is justified.

8. Evaluator's quality checklist

✓ Recalculated expectancy and net expectancy.
✓ Checked the implied profit factor against the reported value.
✓ Identified preprocessing leakage and time-series validation concerns.
✓ Identified multiple-testing / selection bias.
✓ Distinguished arithmetic projections from verified realized performance.
✓ Proposed concrete remediation and forward validation.

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